Ch 2.2.3: Intro to classification

Lecture 9 - CMSE 381

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Weds, Sep 18, 2023

Announcements

Lec #	Date			Reading	HW
1	Mon	8/26	Intro / First day stuff / Python Review Pt 1	1	
2	Wed	8/28	What is statistical learning?	2.1	
3	Wed	9/4	Assessing Model Accuracy	2.2.1, 2.2.2	
4	Fri	9/6	Linear Regression	3.1	HW #1 Due Sun 9/8
5	Mon	9/9	More Linear Regression	3.1	
6	Wed	9/11	Multi-linear regression	3.2	
7	Fri	9/13	Probably more linear regression	3.3	Hw #2 Due Dun 9/15
8	Mon	9/16	Last of the linear regression		
9	Wed	9/18	Intro to classification, Bayes classifier, KNN classifier	2.2.3	
10	Fri	9/20	Logistic Regression	4.1, 4.2, 4.3.1-3	Hw #3 Due Sun 9/22
11	Mon	9/23	Multiple Logistic Regression / Multinomial Logistic Regression /Project day	4.3.4-5	
	Wed	9/25	***Review***		
	Fri	9/27	**Midterm #1**		

Last Time:

- Finished Linear Regression

Announcements:

- Homework #3 Due Sunday Sep 22
- Next Wednesday - Review day
  - Nothing prepped
  - Bring your questions
- Friday 9/27 - Exam #1
  - Bring 8.5x11 sheet of paper
  - Handwritten both sides
  - Anything you want on it, but must be your work
  - You will turn it in

Covered in this lecture

- Ch 2.2.3
- Error rate (classification)
- Bayes Classifier
- K-NN classification

Section 1

Classification Overview

What is classification

Classification: When the response variable is qualitative

$\llcorner Y$

$\hookrightarrow$ list of stuff

Unordered set.

Eye color blue, green, brown
Yellow

Yes / No

- Given feature vector X and qualitative response Y in the set S, the goal is to find a function (classifier) $C(X)$ taking X as input and predicting its value for Y.
- We are more interested in estimating the probabilities that X belongs to each category

Some examples

- Predict whether a COVID19 vaccine will work on a patient given patient's age

Input $x = \text{age}$

Output $y \in \{\text{Works}, \text{Doesn't work}\}$

- An online banking service wants to determine whether a transaction being performed is fraudulent on the basis of the user's IP address, past transactions, etc.

Output $y \in \{\text{Fraud}, \text{Not Fraud}\}$

Inputs

Section 2

Ch 2.2.3: Classification

Error rate

- Training data: $\{(x_1, y_1), \cdots, (x_n, y_n)\}$ with y_i qualitative

  input could have $(x_{1n}, x_{2n}, \ldots, x_{pn})$
- Estimate $\hat{y} = \hat{f}(x)$
- Indicator variable

$$I(y_i \neq \hat{y}_i) = \begin{cases} 1 & \text{if } y_i \text{ and } \hat{y}_i \text{ not the same} \\ 0 & \text{else} \end{cases}$$

actual event

predicted event

Training error rate: Based on data trained model

$$\% \text{ error} = \boxed{\frac{1}{n}\sum_{i=1}^{n} \underbrace{I(y_i \neq \hat{y}_i)}_{\substack{1 \text{ if wrong} \\ 0 \text{ if right}}}}$$

# of incorrect predictions

Test error rate:

$$\mathrm{Ave}(I(y_0 \neq \hat{y}_0))$$

y_0 unseen data point

How often I'm wrong on unseen data

Goal: Find classifiers with minimal error rate.

Best ever classifier

We can't have nice things

Bayes Classifier:
Give every observation the highest probability class given its predictor variables

$$\Pr(Y = j \mid X = x_0)$$

(If I had perfect knowledge of the distribution)

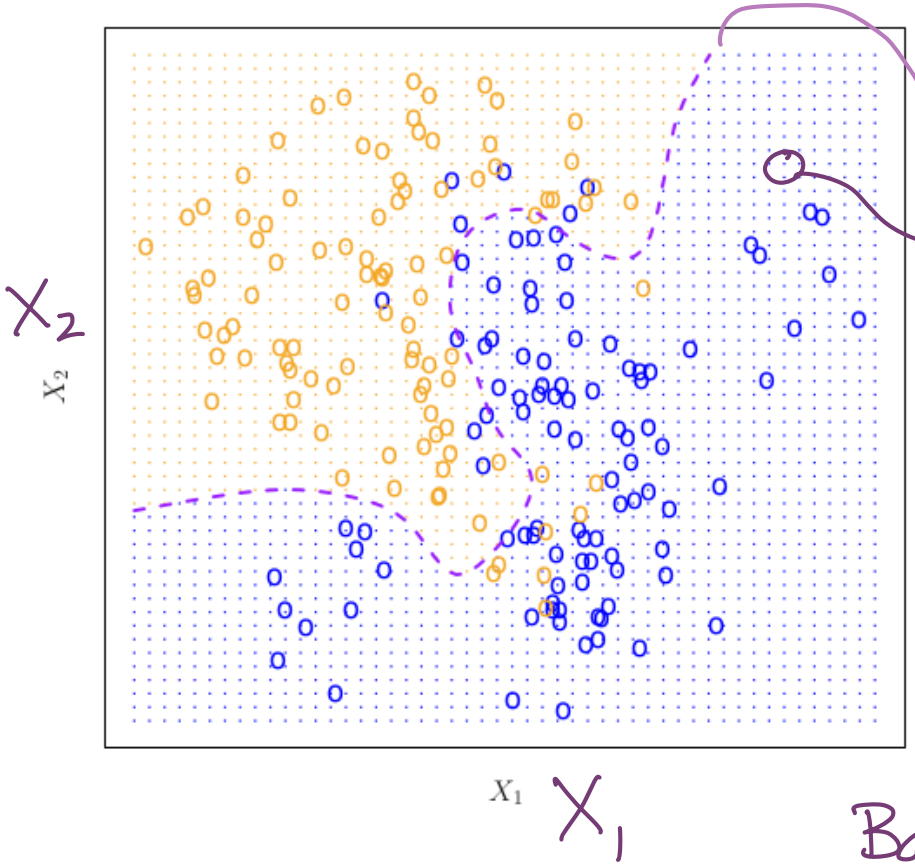
- It is possible to show that this classifier minimizes the test error over all classifiers

An example

- Survey students for amount of programming experience, and current GPA
- Try to predict if they will pass CMSE 381.
- If we have a survey of all students that could ever exist, we can determine the probability of failure given combo of those features.

No single classifier will be 100% correct because students with same input might pass or fail.

Bayes decision boundary



$Y \in \{\text{yellow}, \text{blue}\}$

Dataset : Big circles

Small dot : Blue if higher chance of (x_1, x_2) input having $Y = \text{blue}$

Purple dashed line: Bayes decision boundary location where the higher prob. label switches

Bayes classifier : Predict based on which side of decision boundary

Bayes error rate

$(X_1, \ldots, X_p) = (x_{01}, \ldots, x_{0p})$

- Error at $X = x_0$ My Bayes classifier

$$1 - \max_j \Pr(Y = j \mid X = x_0)$$

1 − Pr(most predicted) = Pr(I'm wrong)

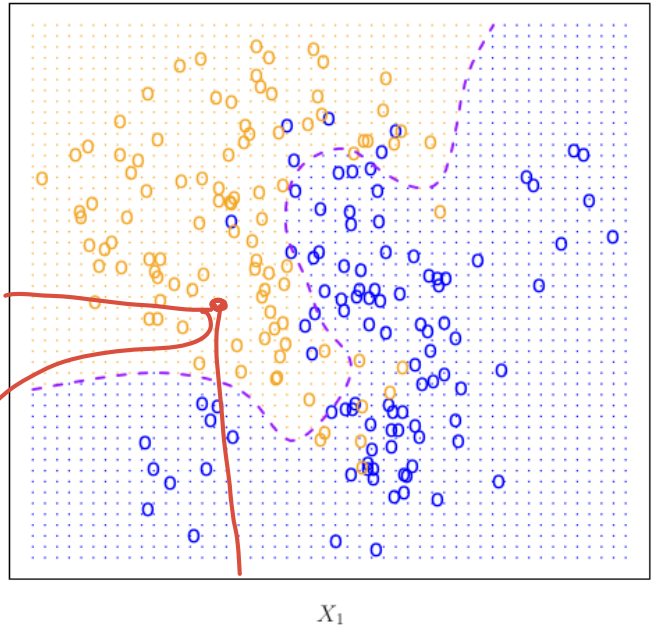
- Overall Bayes error:

$$1 - E\left(\max_j \Pr(Y = j \mid X = x_0)\right)$$

Over all inputs X

E(I'm right)

E(I'm wrong)



This is the irreducable error.

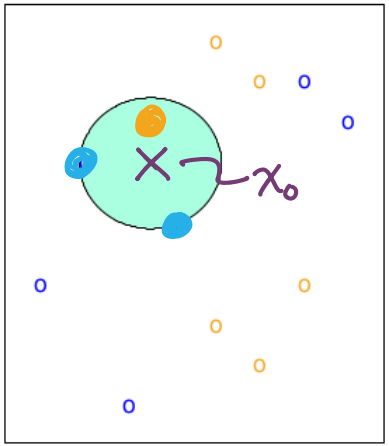
The game

- Goal: find classifier that does as good job as possible estimating the Bayes decision bdy
- Classify based on which side of decision boundary

Section 3

K-Nearest Neighbors Classifier

K-Nearest Neighbors

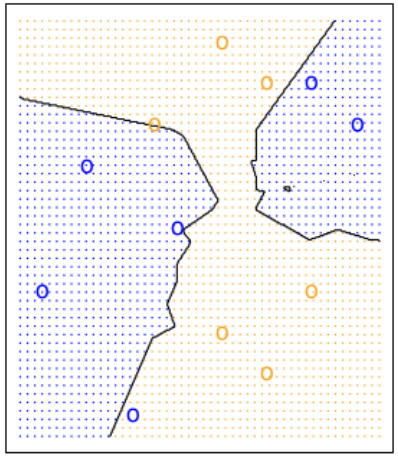


$K = 3$

- Fix K positive integer
- $N(x)$ = the set of K closest neighbors to x
- Estimate conditional proability

$$\Pr(Y = j \mid X = x_0) = \boxed{\frac{1}{K} \sum_{i \in N(x_0)} \mathrm{I}(y_i = j)}$$

- Pick j with highest value



Black line: KNN decision boundary

$P(Y = \text{yellow} \mid X = x_0) = 1/3$

$P(Y = \text{blue} \mid X = x_0) = 2/3$

% of K-NN of x_0 that have label j

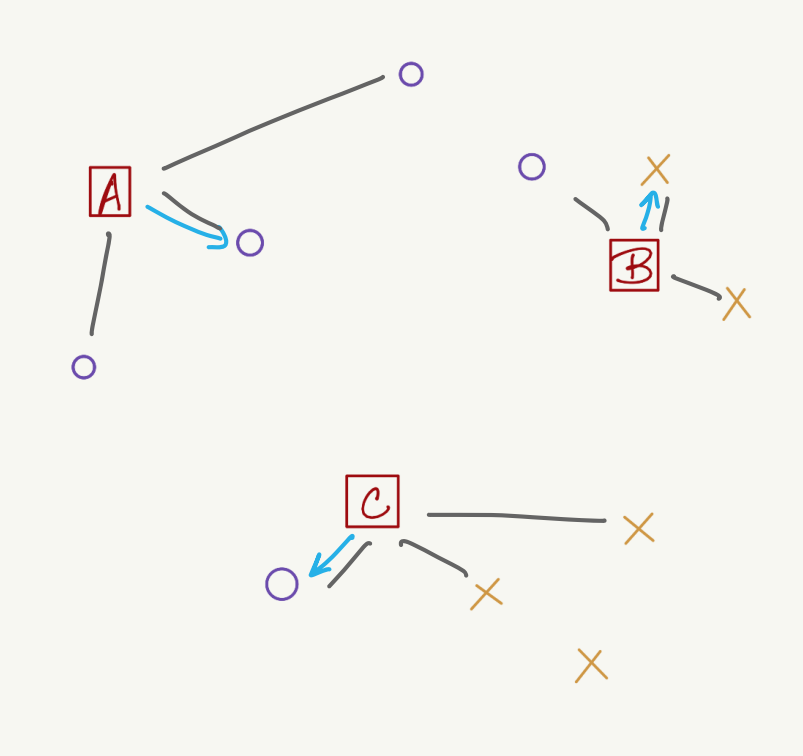
How many of K-nn. to x_0 have label j

$I(y_i = j) = \begin{cases} 1 & \text{obs } i \text{ had } y_i = j \\ 0 & \text{else} \end{cases}$

$\rightarrow$ Predicted val for x_0 is blue.

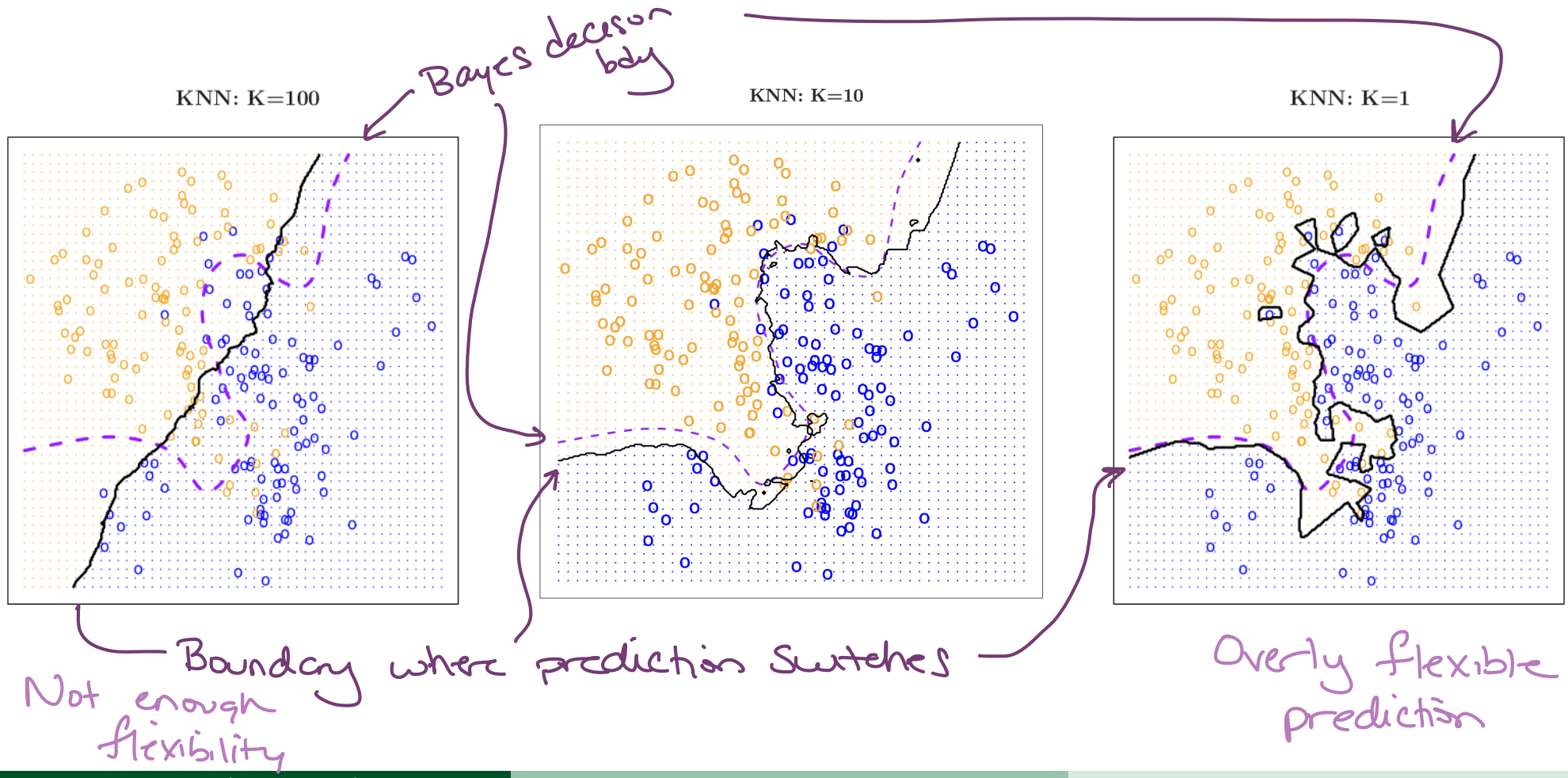
Example

Here label is shown by O vs X. What are the knn predictions for points A, B and C for $k = 1$ or $k = 3$?

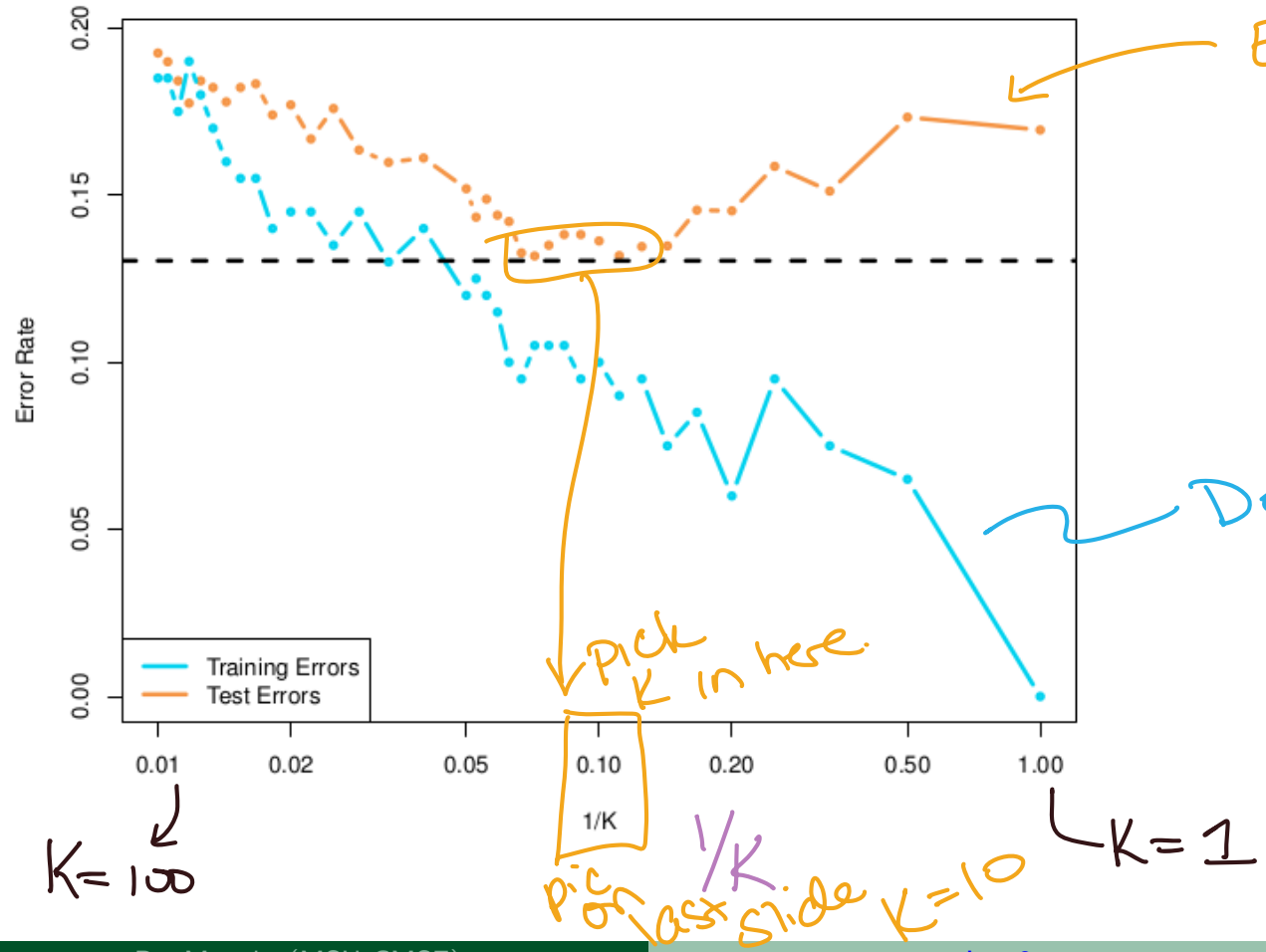


Point	$k = 1$ Prediction	$k = 3$ Prediction
A	O	O
B	X	X
C	O	X

Tradeoff


KNN: K=100
KNN: K=10
KNN: K=1
Bayes decision bdy
Boundary where prediction switches
Not enough flexibility
Overly flexible prediction

More on tradeoff



Error on unseen data

U-shape

Decrease in training error with more flexibility

pick K in here.

pic on last slide

Jupyter notebook

Next time

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